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Thermodynamic assessment of the Gd–Bi and the Ho–Bi systems

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ABSTRACT

The Gd–Bi and the Ho–Bi systems were thermodynamically assessed by CALPHAD (CALculaton of PHAse Diagram) approach based on the available experimental data including thermochemical properties and phase equilibria. The formation enthalpies (at 0 K) of the compounds of the Gd–Bi and the Ho–Bi systems were calculated by the first-principles method and used in the present thermodynamic optimization. The excess Gibbs energies were formulated with the Redlich–Kister polynomial. The intermetallics, Gd_5Bi_3, Gd_4Bi_3, GdBi, $GdBi_2$, Ho_5Bi_3 and HoBi, were treated as stoichiometric compounds. A good agreement between the calculated results and the experimental data was obtained.

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1. Introduction

Rare earth (RE) elements are added into the Bi_2Te_3-based thermoelectric materials to increase phonon-scattering, to raise the Seeback coefficient and to improve thermoelectric properties [1–5]. Therefore the RE-Bi systems deserve to be studied in order to accomplish the RE-Bi-Te thermodynamic database for developing high-performance thermoelectric systems. The calculation of phase diagrams (CALPHAD) method, which is a powerful tool to cut down cost and time during development of materials, effectively provides a clear guideline for material design. In order to better understand the interactions of Gd and Ho with Bi and design high performance thermoelectric materials with RE elements, it is important to study the thermochemical properties and the phase equilibria concerning the Gd–Bi and the Ho–Bi systems and to obtain the thermodynamic parameters of these two systems.

As part of the RE-Bi-Te thermodynamic database, this work presents the thermodynamic descriptions of the Gd–Bi and the Ho–Bi binary systems based on the available experimental data and the first-principles calculations.

2. Evaluation of the available information

2.1. The Gd–Bi system

The partial phase diagram of the Gd–Bi binary system was previously measured by Gambino [6]. It covered the composition range 0–60 at% Bi and reported that three intermetallics, Gd_5Bi_3, Gd_4Bi_3 and GdBi, exist in this system. Later, by means of DTA, XRD and metallography, Abulkhaev [7] reinvestigated the phase diagram of the Gd–Bi system and a new phase, $GdBi_2$ was discovered. The Gd_5Bi_3, Gd_4Bi_3 and GdBi phases were formed from the peritectic reactions, $L+Gd_4Bi_3 \leftrightarrow Gd_5Bi_3$ at 1753 K, $L+GdBi \leftrightarrow Gd_4Bi_3$ at 1853 K and $L+GdBi \leftrightarrow GdBi_2$ at 1183 K, respectively, whereas the GdBi phase melted congruently at about 2043 K. In view of the fact that the solubility of each intermetallic compound was very limited, they were treated as stoichiometric compounds in the present work. There were two eutectic reactions, $L \leftrightarrow$ Hcp_A3 (Gd)$+Gd_5Bi_3$ at 1353 K and $L \leftrightarrow$ Rhomb (Bi)$+GdBi_2$ at 542 K. Because the terminal phases in both sides had very narrow homogeneity ranges, they were treated as pure element phases in the present assessment. Furthermore, a polymorphic transformation, Hcp_A1 (Gd)$\leftrightarrow$Bcc_A2 (Gd) at 1508 K, was observed. The details of the invariant reactions in system are shown in Table 1.

Colinet et al. [8] and Takeuchi and Inoue [9] calculated the value of the enthalpy of mixing of the liquid phase at 50 at% Bi. The formation enthalpies of the intermetallic compounds, Gd_5Bi_3, Gd_4Bi_3, and GdBi, were measured in Refs. [10] by calorimetry methods, and were predicted in Ref. [8] by Miedema model.

2.2. The Ho–Bi system

The phase equilibria of the Ho–Bi binary system were previously measured by Abulkhaev [11] based on the differential thermal analysis (DTA) method. It was reported that two intermetallics, Ho_5Bi_3 and HoBi, and two terminal solution phases, Hcp_A3 (Ho) and Rhomb_A7 (Bi), existed in this system. The compound Ho_5Bi_3

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Table 1
Invariant reactions of the Gd–Bi binary system.

Invariant reaction	Reaction type	T (K)	x^{L}_{Bi}	References
L↔Gd_5Bi_3+Hcp_A3 (Gd)	Eutectic	1353 1353	~0.149 0.146	[7] This work
L+Gd_4Bi_3↔Gd_5Bi_3	Peritectic	1753 1753	~0.325 0.32	[7] This work
L+GdBi↔Gd_4Bi_3	Peritectic	1853 1853	~0.36 0.37	[7] This work
L↔GdBi	Congruent	2043 2042	0.5 0.5	[7] This work
L+GdBi↔$GdBi_2$	Peritectic	1183 1183	~0.83 0.82	[7] This work
L↔$GdBi_2$+Rhomb_A7 (Bi)	Eutectic	542 541	~0.985 0.989	[7] This work

Table 2
Invariant reactions of the Ho–Bi binary system.

Invariant reaction	Reaction type	T (K)	x^{L}_{Bi}	References
L↔Ho_5Bi_3+Hcp_A3 (Ho)	Eutectic	1428 1428	~0.18 0.179	[11] This work
L+HoBi↔Ho_5Bi_3	Peritectic	1693 1693	~0.31 0.315	[11] This work
L↔HoBi	Congruent	2048 2048	0.5 0.5	[11] This work
L↔HoBi+Rhomb_A7 (Bi)	Eutectic	544 543.9	~0.99 0.995	[11] This work

was formed by the peritectic reactions, L+HoBi↔Ho_5Bi_3 at 1693 K. The compound HoBi melted congruently at about 2048 K. There were two eutectic reactions, L↔Hcp_A3 (Ho)+Ho_5Bi_3 at 1428 K and L↔Rhomb (Bi)+HoBi at 544 K. The details of the invariant reactions are shown in Table 2.

In addition, the thermochemical data of the Ho–Bi binary system were reported by several groups. The enthalpy of mixing of the liquid phase at 50 at% Bi was calculated in Refs. [8,9]. Colinet et al. [8] predicted the values of the formation enthalpies of the intermetallic compounds Ho_5Bi_3 and HoBi by the Miedema model.

3. Calculation method

3.1. First-principle calculations

In order to supplement the thermochemical properties for intermetallics and have a more reliable description for the thermodynamic parameters, the first-principles method was used to calculate the formation enthalpies of the binary compounds in the Gd–Bi and the Ho–Bi systems. The scalar relativistic all-electron Blöchl's projector augmented-wave (PAW) [12,13] method within the generalized gradient approximation (GGA) was adopted, as implemented in the highly-efficient Vienna ab initio simulation package (VASP) [14,15]. For the GGA exchange-correlation function, the Perdew–Burke–Ernzerhof parameterization (PBE) [16–18] was employed. A plane-wave energy cutoff of 400 eV was held constant for the A_xB_y compounds (A=Gd or Ho, B=Bi, and x and y are the stoichiometric numbers of the elements A and B respectively in the related compound). Brillouin zone integrations were performed using Monkhorst–Pack [19,20] K-point meshes, and the K-point meshes of compounds were $12\times12\times12$. The total energy was converged numerically to less than 1×10^{-6} eV/unit with respect to electronic, ionic and unit cell degrees of freedom. All calculations were performed using the "high" setting within VASP to avoid wrap-around errors. Furthermore, the energy vs. volume equations of state (EOS) [21,22] was used to calculate the minimum of the total energy and lattice constants.

In order to calculate the minimum of total energy of each compound, the volume-dependent energy curves obtained from first-principles calculations are fitted by means of the third-order Birch–Murnaghan equation of state [23–25] given by:

$$E(V)=E_0+\frac{9V_0B_0}{16}\left\{\left[\left(\frac{V_0}{V}\right)^{2/3}-1\right]^3B_0'+\left[\left(\frac{V_0}{V}\right)^{2/3}-1\right]^2\left[6-4\left(\frac{V_0}{V}\right)^{2/3}\right]\right\} \quad (1)$$

where V is the volume of primitive cell; $E(V)$ is the total energy corresponding to V. The equilibrium properties fitted with EOS include the equilibrium volume (V_0), equilibrium energy (E_0), equilibrium modulus (B_0) and its pressure derivative (B_0').

The formation enthalpies of the intermetallic compounds are calculated by the following equation:

$$\Delta H(A_xB_y)=E_{total}(A_xB_y)-xE_{total}(A)-yE_{total}(B) \quad (2)$$

where $E_{total}(A_xB_y)$, $E_{total}(A)$ and $E_{total}(B)$ are the calculated total energies (per atom at $T=0$ K) of the intermetallic compound A_xB_y, pure A and pure B, respectively.

3.2. Thermodynamic modeling

The lattice stabilities of all pure elements are cited from Dinsdale [26]. Different models are employed to describe the liquid phase and intermetallic compounds in the Gd–Bi and the Ho–Bi binary systems.

3.2.1. Liquid phase

The liquid phase is described using a substitutional solution model based on random mixing of the constituent atoms. The Gibbs energy of liquid phase is expressed by the following expression [27]:

$$G_m^{Liq}=x_A{}^0G_A^{Liq}+x_B{}^0G_B^{Liq}+RT(x_A\ln x_A+x_B\ln x_B)+x_Ax_B\sum_{i=0}^{n}{}^iL_{A,B}^{Liq}(x_A-x_B)^i \quad (3)$$

where R is the gas constant, x_i is the mole fraction of the component i, and ${}^iL_{A,B}^{Liq}$ is the interaction parameter between species A and B, which is formulated as a linear function of temperature as ${}^iL_{A,B}^{Liq}=a+bT$, and the coefficients a and b are the parameters to be optimized.

3.2.2. Intermetallic compounds

There are totally six intermetallic compounds in the Gd–Bi and the Ho–Bi binary systems. All these compounds, Gd_5Bi_3, Gd_4Bi_3, GdBi, $GdBi_2$, Ho_5Bi_3 and HoBi, are treated as stoichiometric compounds A_mB_n, since these compounds have not the observed homogeneity ranges in Refs. [7,11]. Meanwhile, due to lack of heat capacities of these phases we estimate these properties according to the Neumann–Kopp rule. The Gibbs energy of the compound A_mB_n per mole of atoms is expressed as follows:

$$G_m^{A_mB_n}=\frac{m}{m+n}{}^0G_A^{SER}+\frac{n}{m+n}{}^0G_B^{SER}+c+dT \quad (4)$$

where the parameters c and d are to be optimized.

4. Results and discussion

4.1. First-principles calculations

The Birch–Murnaghan equation of state (Eq. (1)) is adopted in the present work to determine the minimum of total energy and the lattice constants. The experimental lattice parameters are used as the starting point to perform the structural calculations for Gd_4Bi_3, Gd_5Bi_3, GdBi, Ho_5Bi_3 and HoBi [28–30]. The total energies of these four compounds are optimized for different c/a and b/a ratios around the equilibrium cells. And these two equilibrium parameters, c/a and b/a, are fixed by calculating the equilibrium volume using Birch–Murnaghan equation of state (Eq. (1)). In Table 3 the optimized results of the lattice constants a, b and c are summarized and compared with available experimental results taken from Refs. [28–30]. It is noticed that the present theoretical lattice constants are consistent with the experimental values, within a difference less than 1%. The equilibrium volume (V_0), equilibrium energy (E_0), equilibrium modulus (B_0) and its pressure derivative (B_0') fitted by EOS are listed in Table 4.

Based on the minimum of total energy by Eqs. (1) and (2), the formation enthalpies of the binary compounds, Gd_5Bi_3, Gd_4Bi_3, GdBi, Ho_5Bi_3 and HoBi, are calculated. The comparison between the calculated results and the experimental data is listed in Table 5, together with Miedema model's results. For the Gd–Bi system, the results of the first-principles calculations of formation enthalpy of GdBi is consistent with the datum reported by Borzone et al. [10], with the relative deviation within 6.45%. And for the formation enthalpy of Gd_4Bi_3 and Gd_5Bi_3 compounds, the relative deviations are about 14.9% and 15% respectively. It indicates that the present first-principles calculations are reasonable and acceptable. Therefore, the first-principles calculations' results of the Ho–Bi binary system are adopted in the present assessment due to lack of the formation enthalpy data of the related compounds.

The experimental formation enthalpy of $GdBi_2$ is not reported in literature and the total energy of $GdBi_2$ compound cannot be calculated as well by First-principles method due to lack of the experimental lattice data. Therefore, the formation enthalpy of $GdBi_2$ compound is calculated by first-principles method using the structures of $REBi_2$ and $RESb_2$. By using the structures of $EuBi_2$ [31], $EuSb_2$ [32], $GdSb_2-1$ [33], $GdSb_2-2$ [33] and $YbSb_2$ [34], the formation enthalpy of $GdBi_2$ are calculated to be −54031.64, −56876.94, −21531.95, −53812 and −56839.61 J/mol respectively, which are more positive than the value predicted by Miedema model in literature [8] and the relative deviations exceed 15%. Considering these deviations and the values calculated by Miedema model are much more positive than the experimental data of other compounds in the Gd–Bi system, the above calculated formation enthalpies of $GdBi_2$ are not adopted in present assessment because apparently the correct structure data have not been found. And the assessment work continues based on the Miedema model value and the phase boundaries for the $GdBi_2$ compound.

4.2. Thermodynamic assessment

With the lattice stabilities of all pure elements cited from Dinsdale [26] and the PARROT module of the Thermo-Calc software [35], the phase relations in the Gd–Bi and the Ho–Bi binary systems are carefully assessed based on the experimental and theoretical data of the phase diagram and thermochemical properties. Tables 6 and 7 list the optimized thermodynamic parameters for the Gd–Bi and the Ho–Bi systems, respectively.

Table 3
The lattice constant a, b and c compared to other available experimental data.

Phase	Pearson symbol	Space group	a (Å)	B (Å)	C (Å)	References
Gd_5Bi_3	hP16	$P6_3/mcm$	9.252	–	6.412	First-principle
			9.182	–	6.426	Experiment [28]
Gd_4Bi_3	cI28	$I\bar{4}3d$	9.4418	–	–	First-principle
			9.3853	–	–	Experiment [29]
GdBi	cF8	$Fm\bar{3}m$	6.3562	–	–	First-principle
			6.3108	–	–	Experiment [29]
Ho_5Bi_3	oP32	Pnma	8.194	9.469	11.942	First-principle
			8.128	9.385	11.874	Experiment [30]
HoBi	cF8	$Fm\bar{3}m$	6.2853	–	–	First-principle
			6.2291	–	–	Experiment [29]

Table 4
The calculated equilibrium energy (E_0), equilibrium volume (V_0), equilibrium modulus (B_0) and its pressure derivative (B_0').

Phase	E_0 (eV)	V_0 (Å^3)	B_0 (GPa)	B_0'
Gd_5Bi_3	−82.703	475.316	59.431	4.384
Gd_4Bi_3	−72.05	420.857	57.257	4.317
GdBi	−10.532	64.199	53.638	4.234
Ho_5Bi_3	−161.917	926.568	58.104	4.259
HoBi	−10.332	62.075	54.837	4.213

Table 5
The formation enthalpies at 0 K obtained by the first-principles calculations, the Miedema model [2] and reported experimental data [3].

Phase	Prototype	Experimental (J/mol)	First-principles in this work (J/mol)	Miedema model (J/mol)
Gd_5Bi_3	Mn_5Si_3	−87,400 [10]	−74,250.21	−76,000 [8]
Gd_4Bi_3	Th_3P_4	−92,000 [10]	−78,210.1	−81,000 [8]
GdBi	NaCl	−96,700 [10]	−90,462.48	−83,000 [8]
$GdBi_2(EuBi_2)$	$HfGa_2$	–	−54,031.64	−67,000 [8]
$GdBi_2(EuSb_2)$	$CaSb_2$	–	−56,875.94	−67,000 [8]
$GdBi_2(GdSb_2–1)$	$HoSb_2$	–	−21,531.95	−67,000 [8]
$GdBi_2(GdSb_2–2)$	$SmSb_2$	–	−53,812	−67,000 [8]
$GdBi_2(YbSb_2)$	$ZrSi_2$	–	−56,839.61	−67,000 [8]
Ho_5Bi_3	Er_5Bi_3	–	−67,807.03	−71,000 [8]
HoBi	NaCl	–	−84,446.2	−77,000 [8]

Table 6
Thermodynamic parameters of the Gd–Bi binary system obtained in this work.

Phase	Thermodynamic parameters
Liquid (Gd, Bi)	${}^0L^{Liq}_{Gd,Bi} = -200{,}000-2.0294T$
	${}^1L^{Liq}_{Gd,Bi} = -3168.5926+4.7319T$
	${}^2L^{Liq}_{Gd,Bi} = 38{,}200+4.37T$
	${}^3L^{Liq}_{Gd,Bi} = -30{,}000+2.8846T$
Gd_5Bi_3	$G^{Gd_5Bi_3}_{Gd:Bi} = -75{,}332.61+4.1T+0.625GHSERGD+0.375GHSERBI$
Gd_4Bi_3	$G^{Gd_4Bi_3}_{Gd:Bi} = -83{,}701.2+5.1T+0.5714GHSERGD+0.4286GHSERBI$
GdBi	$G^{GdBi}_{Gd:Bi} = -88{,}053.58+3.8T+0.5GHSERGD+0.5GHSERBI$
$GdBi_2$	$G^{GdBi_2}_{Gd:Bi} = -68{,}103.64+4.84T+0.333GHSERGD+0.667GHSERBI$

Table 7
Thermodynamic parameters of the Ho–Bi binary system obtained in this work.

Phase	Thermodynamic parameters
Liquid (Ho, Bi)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ho,Bi}} = -190{,}000 + 0.57036T$
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ho,Bi}} = -1000 + 4.2493T$
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ho,Bi}} = 50{,}200 + 4.4783T$
	${}^{3}L^{\mathrm{Liq}}_{\mathrm{Ho,Bi}} = -18{,}001.0647 + 3.2102T$
Hcp_A3 (Ho)	${}^{0}L^{\mathrm{Hcp_A3(Ho)}}_{\mathrm{Ho,Bi}} = -117{,}710.88$
Ho_5Bi_3	$G^{\mathrm{Ho_5Bi_3}}_{\mathrm{Ho,Bi}} = -72{,}329.6 + 5.7T + 0.625GHSERHO + 0.375GHSERBI$
HoBi	$G^{\mathrm{HoBi}}_{\mathrm{Ho:Bi}} = -85{,}973.7 + 4.9T + 0.5GHSERHO + 0.5GHSERBI$

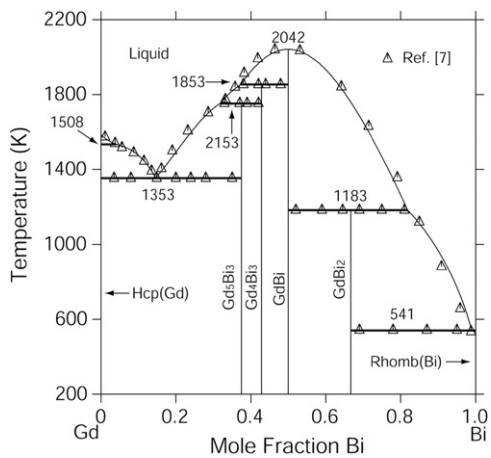

Fig. 1. The assessed Gd–Bi phase diagram in comparison with the experimental data [7].

As shown in the next sections, most of the thermodynamic and phase boundary data are well reproduced.

4.2.1. The Gd–Bi binary system

Fig. 1 shows the calculated phase diagram of the Gd–Bi system together with the experimental data taken from Ref. [7]. It can be seen that the calculated phase diagram is in a reasonable agreement with the experimental data. The calculated peritectic temperatures of the intermetallic compounds, Gd_5Bi_3, Gd_4Bi_3, and $GdBi_2$, are 1753, 1853 and 1183 K, respectively, and the congruent melting temperature of GdBi is 2042 K. The eutectic reaction temperatures for L↔Fcc_A1 (Gd)+Gd_5Bi_3 and L↔Rhomb (Bi)+$GdBi_2$ are 1353 and 541 K, respectively. The details of the invariant reactions in this binary system together with the data from literatures are listed in Table 1 for comparison.

Fig. 2 presents the calculated mixing enthalpies of the liquid phase at 2500 K, together with the results reported by Refs. [8,9], from which it can be seen that a good agreement is obtained. Fig. 3 represents the assessed formation enthalpies of all the intermetallic compounds at 300 K together with the experiment data [10] and the first-principles calculations. The assessed values are more close to the present first-principles calculations than the data reported by Borzone et al. [10], even though the relative deviation between the latter two is not very large.

4.2.2. The Ho–Bi binary system

The calculated phase diagram of the Ho–Bi system is shown in Fig. 4 together with the experimental data from Abulkhaev [11]. The peritectic temperature of the intermetallic compound Ho_5Bi_3 is 1693 K, and the congruent melting temperature of HoBi is

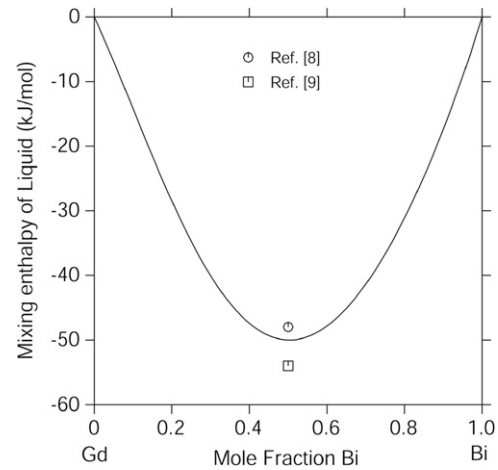

Fig. 2. The mixing enthalpy of liquid Gd–Bi alloys at 2500 K: comparison between calculated values and reported data [8,9]. Reference states are liquid Gd and liquid Bi.

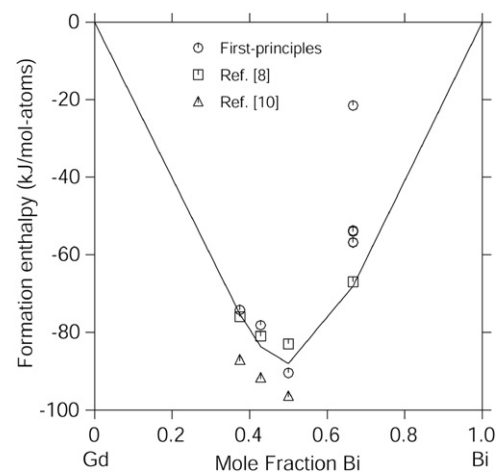

Fig. 3. The formation enthalpy of Gd–Bi intermetallic compounds at 300 K: comparison between calculated values with reported data [8,10]. Reference states are Hcp (Gd) and Rhomb (Bi).

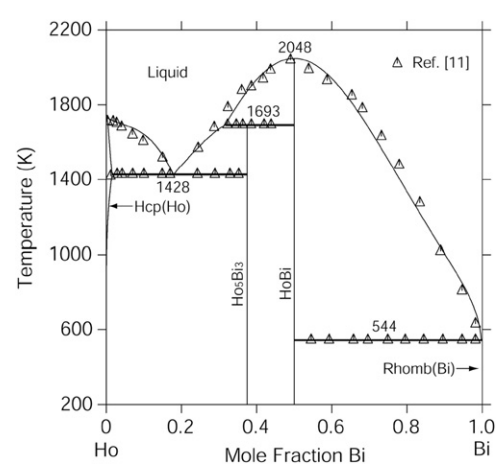

Fig. 4. The calculated Ho–Bi phase diagram in comparison with the experimental data [11].

2048 K. The eutectic reaction temperatures for L↔Hcp_A3 (Ho)+Ho_5Bi_3 and L↔Rhomb (Bi)+HoBi are 1428 and 543.9 K, respectively. All the invariant reactions in the Ho–Bi system are

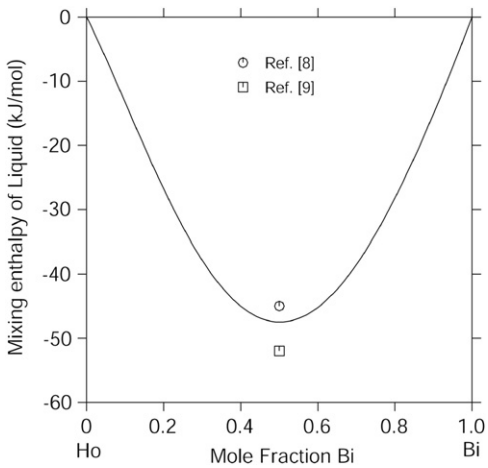

Fig. 5. The mixing enthalpy of liquid Ho–Bi alloys at 2500 K: comparison between calculated values and reported data [8,9]. Reference states are liquid Ho and liquid Bi.

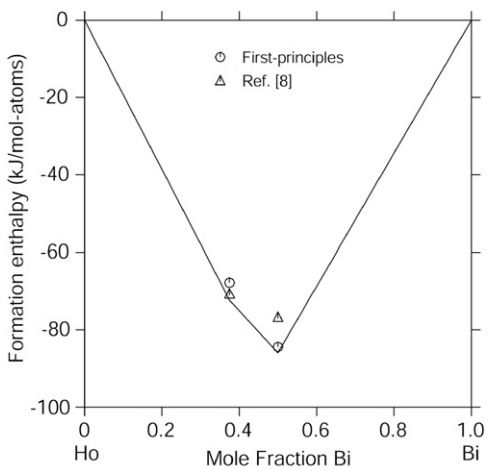

Fig. 6. The formation enthalpy of Ho–Bi intermetallic compounds at 298 K: comparison between calculated values with reported data [8]. Reference states are Hcp (Ho) and Rhomb (Bi).

listed in Table 2. The calculated results are in good agreement with most of the experimental data [11].

Fig. 5 presents the calculated mixing enthalpies of the liquid phase at 2500 K, together with the results reported by Refs. [8,9], from which it can be seen that a good agreement is obtained. The assessed formation enthalpies of the intermetallic compounds at 298 K are compared with the first-principles calculations, as shown in Fig. 6. A consistent agreement is obtained.

5. Conclusions

With the assistance of the present first-principles calculations and the experimental data of the phase equilibria and the thermochemical properties available in literatures, the Gd–Bi and the Ho–Bi systems are assessed thermodynamically. For each of the systems, a consistent set of thermodynamic parameters has been obtained, and all of the calculated temperatures and compositions of the invariant reactions as well as the thermochemical properties are in good agreement with the reported experimental data.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2013.01.003.

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